PisteChrono PX-8 Timing Loop System – Commissioning Test Report

Wielerbaan Rabot vzw · Rabotpiste indoor velodrome, Gasmeterlaan 212, 9000 Ghent · CTR-PX8-2026-03 rev. C · Issued 24 September 2026

1. SCOPE AND OUTCOME

This report records the commissioning tests of the PisteChrono PX-8 transponder timing system installed under the 250 m timber track of the Rabotpiste. The system replaces the photocell and pressure-strip timing that had been in use since 2009. It was supplied and installed by Verhaeghe Sporttechniek bv of Kortrijk during the summer resurfacing (track closed from 7 July to 31 August 2026).

Tests ran from 15 to 19 September 2026 with 46 club riders and two derny pacers. Acceptance criteria are those of purchase specification `WR-SPEC-2026-02`; a pass requires every loop or sample to meet the criterion. **Result: commissioned with two open items.** Seven of the nine tests passed, CT-07 passed with a remark (Table 2) and CT-04 (detection rate) failed for loop L6 only. The system is released for racing from 3 October 2026, the first Thursday Track Night of the winter season, under the conditions in section 3. Revision C corrects the cable lengths of loops L4 and L5, which were swapped in revision B of 22 September.

2. INSTALLED CONFIGURATION

- Decoder `PX8-DEC-4`, serial `D4-11872`, in timing room T-03 above the home straight; firmware `PX8 FW 5.4.2`, loaded on 12 September 2026 (the factory test in Kortrijk on 26 August used `5.3.9`; 5.4.2 adds the derny filter used in test CT-09).
- Transponders: 120 active units type `PXT-12`, fitted with a clip to the right-hand fork blade, 22 to 28 cm above the track; battery life stated as 3 years.
- The decoder and the photo-finish camera (10 000 lines per second) share a GPS-disciplined master clock.
- Loops are numbered in race direction (anticlockwise) and positions are measured along the measurement line, 20 cm from the inner edge of the track, starting at the finish line.

TABLE 1. LOOP POSITIONS AND PER-LOOP DETECTION RESULTS (ALL SAT PASSES)

Loop	Track line served	Position (m)	Cable (m)	Decoder port	Passes	Detection rate	Result
L1	Finish line (lap and finish times)	0.0	31	A1	4 912	99.98 %	Pass
L2	200 m line (flying 200 m start)	50.0	58	A2	4 907	99.94 %	Pass
L3	Pursuit line B (back straight)	103.0	84	A3	4 905	99.90 %	Pass
L4	100 m line	150.0	96	A4	4 903	99.88 %	Pass
L5	Bend 3 exit split	190.0	72	B1	4 900	99.86 %	Pass
L6	Pursuit line A (home straight)	228.0	39	B2	4 898	99.12 %	Fail
L7	Back-up finish loop	0.6	32	B3	4 912	99.96 %	Pass
L8	Infield warm-up lane exit	n/a	47	B4	1 236	99.84 %	Pass

3. CONDITIONS FOR USE

- Decoder profile `RACE-250` for events, `TRAIN-250` for training; minimum lap time filter 12.0 s. The photo-finish decides any dispute closer than 5 ms.
- Riders without a club transponder rent a `PXT-12` at the front desk for EUR 5.00 per session (deposit EUR 40.00); until 30 September 2026 the rental was EUR 3.50.

Fictional document created for testing a question-answering system. All names, figures and organisations are invented.

4. TEST PROGRAMME AND ACCEPTANCE CRITERIA

TABLE 2. COMMISSIONING TESTS CT-01 TO CT-09

Test	Description	Criterion	Measured	Result
CT-01	Loop insulation resistance at 500 V DC	at least 20 MΩ per loop	min. 180 MΩ (L6)	Pass
CT-02	Loop inductance after tuning	80 to 120 µH	94 to 108 µH	Pass
CT-03	Clock drift vs photo-finish over 4 h	at most 0.5 ms	0.21 ms	Pass
CT-04	Detection rate per loop (Table 1)	at least 99.80 %	99.12 % (L6)	Fail
CT-05	Finish time deviation at L1 vs photo-finish	95th percentile at most 2.0 ms	1.3 ms	Pass
CT-06	Bunch passes, wheels 0.5 m apart	no swapped rider IDs	0 swaps / 600 passes	Pass
CT-07	Scoreboard update latency	at most 300 ms	280 ms (peak 340 ms)	Remark
CT-08	UPS run time on mains failure	at least 30 min	42 min	Pass
CT-09	Derny pacing: pacer does not mask rider	100 % of riders read	100 % (318 passes)	Pass

5. SPEED SWEEP AT THE FINISH AND PURSUIT LOOPS

Speed at the line is computed from the L1 to L7 time. L6 misses grow with speed: the loop sits too deep under panels 141 and 142.

TABLE 3. MISSED READS AND FINISH DEVIATION BY SPEED BAND, 15 TO 19 SEPTEMBER 2026 (DEV. = DEVIATION FROM PHOTO-FINISH, P95 = 95TH PERCENTILE)

Speed band	Typical use	Passes	Missed L1	Missed L6	L1 dev. p95	L1 dev. max
25–35 km/h	Warm-up, youth groups	1 118	0	2	0.9 ms	1.6 ms
35–45 km/h	Endurance, scratch	1 386	0	5	1.0 ms	1.9 ms
45–55 km/h	Bunch finishes, madison	1 052	1	9	1.2 ms	2.4 ms
50–60 km/h	Derny-paced (CT-09)	318	0	4	1.3 ms	2.2 ms
55–65 km/h	Sprint, keirin	684	0	12	1.5 ms	2.8 ms
65–75 km/h	Flying 200 m, elite	354	0	11	1.8 ms	3.1 ms
All bands	Total on track	4 912	1	43	1.3 ms	3.1 ms

6. OPEN ITEMS

- **P-01 – Loop L6 detection (CT-04 fail).** Lift panels 141 and 142, reseat the loop 18 mm below the riding surface and re-tune. Responsible: Bram Verhaeghe, supplier site engineer. Track closed on 29 September 2026, 07:00 to 16:00; CT-04 retest for L6 (at least 1 000 passes) by 10 October 2026.
- **P-02 – Scoreboard latency peaks (CT-07 remark).** Move the results export to the end of each race. Responsible: Jolien Van Acker, technical manager of Wielerbaan Rabot vzw, by 3 October 2026.

Warning. Until P-01 is closed, times from loop L6 are not official; pursuit finish times come from loop L3 or the photo-finish.

Sign-off. Jolien Van Acker (Wielerbaan Rabot vzw), Bram Verhaeghe (Verhaeghe Sporttechniek bv) and chief timekeeper Sofie Claeys signed on 24 September 2026.

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